Human-Scene Reconstruction: A Review of Representations, Methods, and Embodied Applications

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Abstract

Human-scene reconstruction aims to recover people and their surrounding environments in a shared 3D or 4D world representation. The problem is broader than human mesh recovery and broader than scene reconstruction alone: a useful system must estimate cameras, scene geometry, human bodies, identities, trajectories, contact, and metric scale in a mutually consistent coordinate frame. This review summarizes the field through a compact bibliography of fewer than fifteen references. We organize the discussion around standard review categories: problem formulation, representations, datasets and supervision, methodological families, interaction and dynamics, embodied applications, and open challenges. Within this structure, MonoFusion is used as an example of sparse-view dynamic reconstruction, CRISP as an example of contact-guided simulation-ready reconstruction, and EmbodMocap as an embodied-agent data and evaluation perspective.

1 Introduction

Humans do not move in isolation. Walking assumes a ground plane, sitting assumes support, reaching depends on object placement, and climbing depends on stairs or footholds. At the same time, the environment helps explain the human: floors constrain root translation, contact constrains body pose, static background geometry constrains camera motion, and occluders explain depth ordering. Human-scene reconstruction studies this coupled inference problem. Its goal is to recover not only a plausible human body, and not only a plausible 3D scene, but a coherent world in which the body, camera, and environment agree.

The field has grown out of several neighboring areas. Parametric body models make human shape and pose optimizable [1, 2]. Scene-aware datasets show that pose estimation must be evaluated against physical context [3, 4]. World-grounded human motion methods address the ambiguity between human motion and camera motion [5, 6]. Feed-forward geometry models provide strong priors for cameras and scene structure [7, 8]. Recent work expands the scope toward sparse-view dynamic reconstruction, real-to-sim contact recovery, and embodied-agent datasets [9–11].

This review keeps the title and section structure centered on the general problem of human-scene reconstruction. Individual papers are used as representative examples inside broader categories, rather than as section titles. The bibliography is deliberately compact, capped at fifteen references, to emphasize the conceptual lineage instead of exhaustive coverage.

2 Problem Formulation

A common input is a set of images or videos

$$\mathcal{I} = \{I_{v,t}\}_{v=1,t=1}^{V,T}, \tag{1}$$

where v indexes camera view and t indexes time. A human-scene reconstruction method typically estimates camera intrinsics and extrinsics $\{K_{v,t}, T_{v,t}\}$, scene geometry $\mathcal{S}$, human states $\mathcal{H} = \{\theta_{i,t}, \beta_i, R_{i,t}, x_{i,t}\}$, identity associations $a_{i,v,t}$, and sometimes contact variables $c_{i,t}$ or visibility masks. The human parameters usually include pose θ, shape β, global orientation R, and global translation x under a body model such as SMPL or SMPL-X [1, 2].

The defining requirement is consistency. A reconstruction should be geometrically consistent across views, temporally consistent across frames, metrically consistent between humans and scenes, and physically consistent with contact and free space. These requirements make the task substantially harder than per-frame human mesh recovery. A body can have accurate 2D reprojection while floating above the floor; a scene point cloud can look plausible while lacking support surfaces; a camera trajectory can be visually acceptable while producing the wrong human scale. Human-scene reconstruction asks these estimates to agree in one world frame.

The output can be used at several levels of fidelity. For visualization, it may be enough to recover cameras, human meshes, and coarse geometry. For AR/VR, occlusion, metric scale, and temporal stability become more important. For robotics and embodied AI, the output must also support collision, contact, imitation, and control. This application spectrum explains why recent papers increasingly discuss contact, physical plausibility, and downstream policy learning rather than only pose or rendering accuracy.

3 Representations

3.1 Human representation

Parametric human models remain the dominant human representation because they provide a low-dimensional, differentiable, and temporally trackable surface. SMPL parameterizes a skinned body mesh with pose-dependent deformations [1]. SMPL-X extends the representation to full-body capture with hands and face [2]. These models give reconstruction systems a strong prior over body shape and articulation, and they make it possible to optimize or regress human pose from images.

However, the same compactness also creates limitations. Parametric bodies do not directly encode clothing, carried objects, deformation, or detailed contact. They also do not by themselves determine global scale, floor height, or scene layout. In human-scene reconstruction, the body model is therefore best understood as one component in a larger world model: it supplies articulated human geometry, while cameras, scene geometry, identity tracking, and contact reasoning supply context.

3.2 Scene representation

Scene representations vary from sparse point clouds and dense point maps to meshes, neural fields, planes, and simulation primitives. Sparse or dense points are flexible and easy to predict with modern feed-forward geometry models [7, 8]. Meshes and implicit surfaces are better suited for rendering and collision. Planar or convex primitives are less photorealistic but more useful for physical simulation, especially when the goal is to infer support surfaces, obstacles, stairs, or furniture contact.

A central design question is whether the scene representation is intended to be visually accurate, metrically accurate, interaction-aware, or simulation-ready. These goals are related but not identical. A visually detailed point cloud may contain noise that makes physics impossible, while a simplified convex scene may omit texture but preserve the support surfaces that matter for behavior. Human-scene reconstruction often needs both levels: visual geometry for perception and functional geometry for interaction.

3.3 Contact and interaction representation

Contact is the main bridge between humans and scenes. It can be represented as binary vertex contact, dense body-scene correspondences, support planes, object attachments, or contact windows over time. Scene-aware datasets such as PROX and RICH made contact and scene constraints central to evaluation [3, 4]. More recent embodied formulations use contact as a signal for physical feasibility and downstream control [10, 11].

The hardest cases are not static contact alone, but changing contact: stepping, sitting, grasping, carrying, climbing, and manipulation. These events reveal which parts of the scene are functionally important. A staircase is not only geometry; it is a sequence of footholds. A chair is not only a mesh; it is a support surface. A carried object is not only an appearance cue; it changes the person's center of mass and feasible motion. A mature human-scene representation must capture these functional relationships.

4 Datasets and Supervision

Human-scene reconstruction depends heavily on the type of supervision available. Studio motion capture provides accurate bodies but weak scene diversity. RGB-D capture gives metric scene geometry but may be limited to indoor settings. Multi-view capture improves 3D reliability but is expensive and less scalable. In-the-wild monocular video provides scale and diversity, but it lacks direct 3D supervision.

PROX introduced a useful scene-constrained setting by fitting human bodies in scanned indoor environments [3]. RICH extended this direction with dense full-body human-scene contact [4]. EMDB provided global 3D human pose and shape in the wild, making world-space evaluation more practical [12]. EmbodMocap further shifts the discussion toward portable in-the-wild 4D capture for embodied agents [11]. Together, these datasets show a progression from body fitting, to scene-aware fitting, to global motion, to embodied interaction data.

Supervision is also becoming more hybrid. Modern systems may combine ground-truth capture, pseudo-labels from geometry models, multi-view consistency, temporal smoothness, contact losses, and physical constraints. This hybrid supervision is necessary because no single dataset provides perfect labels for all variables: cameras, humans, scenes, identities, contacts, dynamics, and physical parameters.

5 Methodological Families

5.1 Scene-constrained optimization

A classical approach is to estimate humans and scenes separately, then refine the human body with scene constraints. This family is conceptually simple and often interpretable: the scene supplies collision, contact, support, or visibility constraints, while the body model supplies pose and shape priors. PROX is a representative example, showing that 3D scenes can reduce pose ambiguities in indoor human reconstruction [3]. RICH strengthens this direction by making dense contact a core signal [4].

The advantage of this family is controllability. One can write explicit losses for penetration, foot-floor contact, temporal smoothness, or body-scene distance. The disadvantage is fragility. Multi-stage optimization depends on initialization, calibrated scenes, accurate masks, and carefully weighted losses. Errors in one module can propagate to the next. The field has therefore moved toward learned priors

Method family	Representative references	Main contribution to human-scene reconstruction
Parametric body recovery	SMPL, SMPL-X [1, 2]	Supplies differentiable human geometry and kinematic priors, but does not solve world grounding alone.
Scene-constrained fitting and data	PROX, RICH, EMDB [3, 4, 12]	Makes global alignment, support, dense contact, and world-space evaluation explicit.
World-coordinate human motion	SLAHMR, WHAM [5, 6]	Addresses camera-human ambiguity and estimates temporally coherent global human trajectories.
Feed-forward scene geometry	DUSt3R, π^3 [7, 8]	Provides image-based camera and scene priors that reduce dependence on slow reconstruction pipelines.
Integrated human-scene reconstruction	Joint reconstruction pipelines	Estimates humans, cameras, and scenes in one coherent world representation.
Dynamic and embodied reconstruction	MonoFusion, CRISP, EmbodMocap [9–11]	Extends the problem toward sparse-view dynamics, contact-guided real-to-sim, and embodied-agent data.

Table 1. A compact taxonomy of human-scene reconstruction with a bibliography capped at fifteen papers. The section structure follows standard review categories; individual systems appear only as representative examples.

and unified architectures that reduce the number of brittle interfaces.

5.2 World-coordinate human motion

Another method family focuses on estimating the human in a stable world coordinate system, especially from moving-camera videos. SLAHMR decouples human and camera motion in videos captured in the wild [5]. WHAM reconstructs world-grounded humans with accurate 3D motion [6]. These methods are important because camera-space accuracy does not guarantee world-space usefulness. For interaction, a person's root trajectory, orientation, and contact timing must be stable in global coordinates.

World-coordinate HMR does not necessarily reconstruct a detailed scene. Nevertheless, it solves an essential subproblem: separating the motion of the observer from the motion of the actor. Without this separation, human-scene reconstruction cannot reliably infer support, contact, or traversed space. In this sense, world-grounded motion is a bridge between classical HMR and full human-scene reconstruction.

5.3 Feed-forward geometry priors

Recent visual geometry models changed the scene side of the problem. DUSt3R predicts dense 3D structure and correspondences from image pairs [7]. π^3 pushes the direction toward scalable feed-forward visual geometry [8]. These models make it possible to infer camera and scene structure without relying entirely on traditional SfM or SLAM.

For human-scene reconstruction, this creates a new design pattern. Instead of reconstructing a scene with a standalone pipeline and fitting humans afterward, one can combine a learned scene prior with a learned human prior and train the system to produce compatible outputs. The remaining challenges are scale alignment, cross-view fusion, temporal stability, identity association, and contact consistency.

5.4 Integrated reconstruction

Integrated methods aim to estimate humans and scenes in a shared representation rather than passing predictions through many loosely connected modules. This design trend is motivated by a practical observation: many failures in human-scene reconstruction come from mismatched coordinate frames, inconsistent scale between human and scene estimates, duplicated identities across views, or per-frame predictions that cannot be fused cleanly.

Integrated systems also make evaluation harder. A method can improve body accuracy while degrading scene quality, or improve camera estimation while producing worse contact. The field therefore needs evaluation that separately reports camera error, human pose and shape, scene accuracy, identity consistency, contact plausibility, and downstream usability.

6 Interaction and Dynamics

6.1 Sparse-view dynamic reconstruction

Real deployments rarely have dense calibrated camera arrays. Sparse-view and monocular settings are therefore essential for practical human-scene reconstruction. MonoFusion addresses sparse-view 4D reconstruction by fusing independent monocular reconstructions from a small number of views into a time- and view-consistent dynamic representation [9]. Its relevance to human-scene reconstruction is that future systems must handle dynamic environments with limited camera coverage, imperfect overlap, and moving people.

This direction also expands the output beyond parametric bodies. Human-scene reconstruction should eventually represent dynamic scene content, moving objects, and non-rigid appearance in addition to SMPL-like humans. Parametric bodies are useful for identity, articulation, and control, but dynamic scene representations are needed for appearance, occlusion, and object motion.

6.2 Contact and physical feasibility

Contact is the main signal that turns a geometric reconstruction into an interaction model. A visually plausible reconstruction can still fail physically if feet slide, bodies penetrate furniture, stairs are not usable as support surfaces, or object motion violates grasp constraints. CRISP makes this point by formulating contact-guided real-to-sim reconstruction with planar scene primitives [10]. The aim is not simply to make a point cloud look right, but to recover geometry that can support simulation and humanoid tracking.

This perspective suggests a stronger evaluation criterion: can the reconstructed world be used? For simulation, the answer depends on support surfaces, collision geometry, contact timing, and object affordances. A simplified planar primitive can be more useful than a noisy dense mesh if it captures the correct physical interface. Conversely, a high-fidelity visual reconstruction can be unusable if it lacks clean contacts or metric scale.

7 Embodied Applications

Embodied applications change the purpose of reconstruction. In a perception-only setting, reconstruction is often judged by geometric error, rendering quality, or pose accuracy. In an embodied setting, the reconstructed world becomes data for training or evaluating agents. EmbodMocap explicitly frames in-the-wild 4D human-scene reconstruction as a resource for embodied agents [11]. It connects capture and reconstruction to downstream monocular reconstruction, physics-based character animation, and robot motion control.

This view introduces three practical requirements. First, the reconstruction must be metric: an agent needs distances, heights, and support surfaces in usable units. Second, the reconstruction must be temporally stable: policies cannot learn from trajectories that jitter or drift. Third, the reconstruction must be physically meaningful: contact, collision, and object motion must be consistent enough to replay or imitate. These requirements explain why embodied work often values clean interaction geometry as much as visual detail.

The embodied perspective also clarifies why different subproblems matter. MonoFusion addresses sparse-view dynamic reconstruction under practical capture limits [9]. CRISP asks whether reconstruction can be converted into simulation-ready contact geometry [10]. EmbodMocap asks whether in-the-wild 4D capture can supply data for embodied agents [11]. These are different points in the same broader review problem: how to recover human-scene worlds that are coherent, dynamic, and useful.

8 Open Challenges

Metric scale and gravity. Absolute scale, floor height, and gravity alignment remain persistent failure modes. Small errors can be visually subtle but physically severe. Future systems should predict scale, gravity, and support planes explicitly rather than treating them as side effects.

Occlusion and missing geometry. Humans often interact with surfaces that are partially hidden: chair seats, stairs, tabletops, handles, and carried objects. Reconstruction systems must infer functional geometry even when it is not fully visible. This is especially important for contact-rich behavior and simulation.

Identity and association. Multi-person multi-view videos require associating bodies across cameras and time. Errors produce duplicated people, identity switches, or inconsistent trajectories. Robust association under occlusion and crowded scenes remains difficult, especially when people interact with shared objects or move through visually similar spaces.

Evaluation beyond geometry. The field needs metrics that evaluate usefulness, not only visual quality. For embodied applications, a reconstruction should be judged by whether it supports simulation, tracking, imitation, navigation, manipulation, or policy learning. This requires benchmarks that combine body accuracy, scene accuracy, contact, dynamics, and downstream control.

9 Conclusion

Human-scene reconstruction is best understood as the recovery of a coherent world state containing humans, cameras, geometry, motion, and interaction. The literature has progressed from parametric body recovery to scene-constrained fitting, world-coordinate human motion, feed-forward geometry priors, integrated human-scene systems, sparse-view dynamic reconstruction, contact-guided real-to-sim, and embodied-agent datasets. The central lesson is that humans and scenes should not be reconstructed as independent objects. They constrain each other through scale, visibility, support, contact, and action. The next stage of the field should therefore move beyond visually plausible 3D outputs toward physically grounded, temporally stable, interaction-aware 4D world models.

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